



Battery degradation prognosis and its health management based on grey iterative modeling

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ABSTRACT

Battery degradation prediction involves inherent uncertainty that poses significant challenges for effective health management. This paper proposes an approach for probabilistic battery capacity prediction and develops a predictive maintenance strategy. First, we construct an iterative multivariable prediction framework with adaptive structure and parameters combined with grey filtering. Given the cumulative nature of capacity degradation, the iterative adjustment enables the model to better adapt to the current health state. The model accounts for internal measurement errors and extends traditional grey models from deterministic to probabilistic prediction. Then, a maintenance strategy is designed that integrates time-based maintenance with condition-based maintenance. Finally, a case study based on real-world datasets validates the effectiveness of the proposed approach. Prediction accuracy and economic performance of the maintenance strategy are both evaluated. Results demonstrate that the enhanced grey modeling approach achieves performance comparable to machine learning methods, which provides significant references for battery risk prevention.

1. Introduction

Batteries offer advantages including high voltage, high energy density, extended lifespan, enhanced safety and environmental compatibility. With the continuous progress of technology, batteries have been widely adopted in many fields, particularly in electric vehicles (EVs) [1]. These serve as alternatives to gasoline fuel to achieve long-range daily commuting for EVs [2]. However, the internal structure of lithium batteries undergoes substantial changes as service cycles accumulate [3]. Although exploring and investigating electrode materials can improve the capacity of the cathode [4], the inevitable loss of ions and active substance consumption during operation still greatly affects the battery long-term performance [5]. Magnesium ion batteries are one of promising alternatives to lithium-ion batteries that enable ultralong cycling life [6]. Continuous use of these batteries still poses severe safety hazards and results in equipment downtime. To mitigate such risks, accurate degradation prediction is a key and challenging task in the battery management system (BMS) [7]. Reasonable and cost-effective maintenance strategies is essential for ensuring high battery utilization rates and economic efficiency.

Battery capacity prediction is a direct way to grasp its degradation [8]. To characterize battery internal state, equivalent circuit models (ECMs) are frequently employed. The parameters of these theoretical models, such as internal resistance and various capacitance values, serve as effective indicators of degradation behavior as they evolve with battery aging [9]. However, capacity is the most direct manifestation. In recent years, numerous scholars have focused on this problem and proposed a number of

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prediction models and algorithms. First, model-based methods are developed. These models focus on degradation mechanisms from the internal electrochemical reaction of the battery and construct a corresponding physical model [10,11]. Degradation process is usually modeled through statistical methods [12]. Wiener process [13] and its extension [14,15], time series [16], and Gaussian process are popular to describe degradation process [17]. Second, data-driven models have been paid attention to prognosis [18]. Signal processing [19,20] and mode decomposition [21] are applied to analyze the degradation signal of the battery. Long short-term memory (LSTM) is applied to estimate the battery capacity and then a hybrid one is proposed to further improve its performance [22,23]. Deep learning methods based on the curve feature are used in prediction [24–27]. Another special data-driven method is grey prediction, which is designed for small-sample modeling scenarios under the grey system theory. Due to its simple modeling process, it is widely used in many fields [28], especially in fault and remaining useful life prediction [29,30]. In scenarios with limited historical data, the grey model reduces the need for model selection to pre-classify degradation modes. Unlike traditional approaches that require expert knowledge to categorize degradation behaviors, the grey model operates through a unified framework that automatically identifies and adapts to various degradation modes. Compared to machine learning, related literature proves the model theoretically and analyzes its relationship with classical time series models, which provides good theoretical interpretations of excellent application effects [31]. Third, more and more hybrid methods are proposed [32]. Filtering, such as Kalman filter (KF) and particle filter (PF) and their extension [33,34] are usually combined with other methods [35]. Grey filtering is a combination of the grey model with PF.

Prediction and maintenance are closely related in prognostics and health management (PHM) [36]. To improve the economy of batteries, research on maintenance strategies has been emphasized. Maintenance decision-making theory represents a comprehensive integration of modern maintenance theory and decision science [37,38]. It is guided by maintenance strategies and focuses on the theoretical and methodological issues related to maintenance decision-making [39,40]. The core is to study the modeling of maintenance targets and the optimization of maintenance parameters [41]. The purpose of maintenance decision research is to comprehensively balance costs and benefits under the premise of reliability and safety. There are three main maintenance methods: corrective maintenance, preventive maintenance and predictive maintenance [42,43]. Predictive maintenance is receiving increasing attention for its proactive approach, which focuses on taking action before a failure occurs [44,45]. This approach relies on making decisions based on the real-time state [46]. With the development of fault prediction technology, more and more machine learning-based maintenance is designed and optimized [47].

However, research gaps remain in battery capacity prediction and health management:

- (1) Most existing literature focuses exclusively on either prognosis or maintenance. While higher prediction accuracy is pursued and maintenance methods become increasingly sophisticated, the integration of these two domains needs further exploration.
- (2) Battery capacity data always has inherent noise for prediction models. It is a derived parameter that must be calculated from current, voltage and other measurable indices. These internal measurement errors are particularly challenging for battery prognostics because capacity prediction relies on integrating these measurements over time.
- (3) Battery degradation exhibits complex nonlinear characteristics and extensive measurement data are universally unavailable. Predicting battery degradation trends from limited data constitutes a critical scientific challenge.

To address these problems, this paper proposes a data-driven prediction method based on grey modeling and presents an innovative approach to battery maintenance management. The comprehensive solution framework is illustrated in Fig. 1. The main objectives of this paper are as follows:

- (1) To account for cumulative capacity degradation effects and internal measurement errors, we construct an iterative grey filtering model with adaptive structure and parameters that enhances grey prediction models through integration with filtering algorithms.
- (2) Within the established framework of time-based and condition-based maintenance strategies, we design a hybrid predictive maintenance strategy based on probabilistic battery capacity prediction results.
- (3) We evaluate the performance of both the prediction model and maintenance strategy through a comprehensive case study. Comparison with other methods validates the competitiveness of the proposed solution for battery prognosis, while sensitivity analysis across different cost scenarios demonstrates its economic viability.

The remainder of this paper is organized as follows. Section 2 states the structure and parameters iteration grey filtering model, construction of the proposed model and approach of parameter estimation, and recommends four kinds of criteria to evaluate accuracy. In Section 3, the predictive maintenance strategy is elaborated in detail with its evaluation method. Subsequently, a case study is carried out in Section 4, including dataset description, capacity prediction, predictive maintenance evaluation and sensitivity analysis. Finally, Section 5 concludes the whole contents and points out some potential research interests.

2. Structure and parameters iteration grey filtering model

Grey modeling is a special data-driven method that can also be classified as a time series model. Through the role of the accumulation operator, it can deal with the system modeling of poor information and limited data.

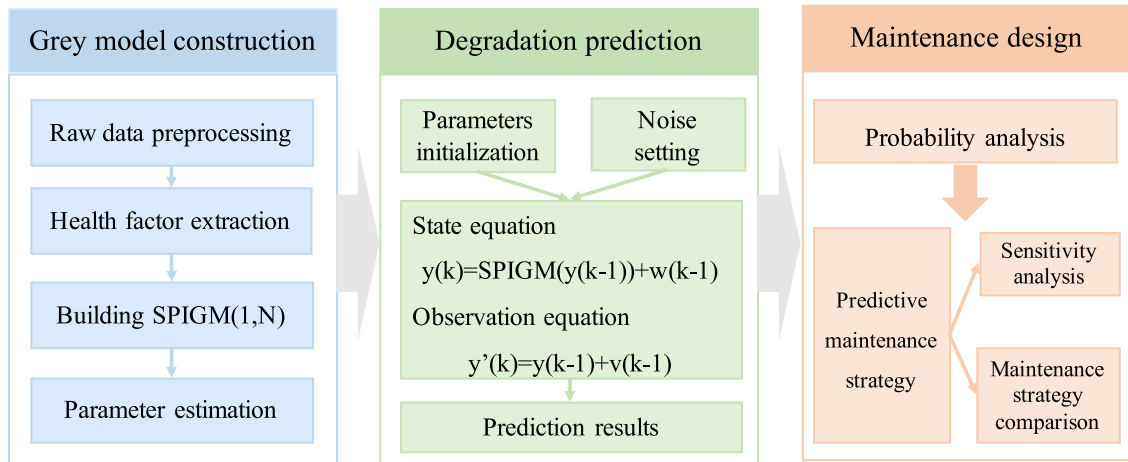


Fig. 1. Solution in this paper.

2.1. Limitation of the classic grey prediction model

Let $Y^{(0)}$ be a system characteristic sequence and $X_i^{(0)}$ ($i = 1, 2, 3, \dots, N$) be the related variables sequence. In the context of battery degradation, the capacity degradation trajectory represents the system characteristic sequence. Various external factors (such as temperature, charge/discharge rates and operational conditions) influencing degradation are treated as related variable sequences. $Y^{(1)}, X_i^{(1)}, i = 1, 2, \dots, N$ are accumulation generated sequences, where

$$y^{(1)}(t_k) = \sum_{j=1}^k h_j y^{(0)}(t_j), x_i^{(1)}(t_k) = \sum_{j=1}^k h_j x_i^{(0)}(t_j), h_k = \begin{cases} 1, k = 1 \\ t_k - t_{k-1}, k = 2, 3, \dots, n \end{cases} \quad (1)$$

The general grey multivariable prediction model framework is based on a differential equation

$$\begin{cases} \frac{dy^{(1)}(t)}{dt} + ay^{(1)}(t) = b_1 x_1^{(1)}(t) + b_2 x_2^{(1)}(t) + \dots + b_N x_N^{(1)}(t) \\ y^{(0)}(t_k) = a \left(\frac{1}{2} (y^{(1)}(t_k) + y^{(1)}(t_{k-1})) \right) + b_1 x_1^{(1)}(t_k) + b_2 x_2^{(1)}(t_k) + \dots + b_N x_N^{(1)}(t_k) \end{cases} \quad (2)$$

Eq. (2) shows the continuous-time and discrete-time forms of grey multivariable model. Modeling based on the accumulated generating operator (AGO) sequence is the most distinctive feature of grey system models. From its discrete form, it can be observed that the current time information depends on the accumulated information about itself and the influencing factors up to that point. This dependency explains the fundamental mechanism and effectiveness of grey system models in predicting time sequences.

As illustrated in Eq. (2), the traditional grey multivariable model constitutes a deterministic and static modeling framework. It performs predictions by a model based on historical data, fixed parameters and subsequently extrapolates future trends. This approach is suitable when future sequences share similar evolutionary characteristics with historical data, or when stable and consistent mappings exist between influencing factors and system behavior sequences. However, when system behavior sequences exhibit multi-stage or phase-varying evolutionary patterns, fixed model structures and deterministic parameters $[a, b_1, b_2, \dots, b_N]$ struggle to effectively capture underlying dynamic information. It leads to diminished prediction accuracy and even model failure. As the complexity of temporal system evolution increases, static models alone become inadequate for extracting essential system dynamics. In response to this limitation, this study proposes an iterative dynamic grey multivariable model to enhance the adaptability and predictive capability under complex and evolving system conditions.

2.2. Construction of proposed model

The differential equation

$$\begin{cases} \frac{dy^{(1)}(t)}{dt} + a(0)y^{(1)}(t) = b_1(0)x_1^{(1)}(t) + b_2(0)x_2^{(1)}(t) + \dots + b_N(0)x_N^{(1)}(t), t = 1, 2, \dots, M \\ \frac{dy^{(1)}(t)}{dt} + a(1)y^{(1)}(t) = b_1(1)x_1^{(1)}(t) + b_2(1)x_2^{(1)}(t) + \dots + b_N(1)x_N^{(1)}(t), t = 1, 2, \dots, M + 1 \\ \vdots \\ \frac{dy^{(1)}(t)}{dt} + a(m)y^{(1)}(t) = b_1(m)x_1^{(1)}(t) + b_2(m)x_2^{(1)}(t) + \dots + b_N(m)x_N^{(1)}(t), t = 1, 2, \dots, M + m \end{cases} \quad (3)$$

is called structure and parameters iteration grey multivariable prediction model, abbreviated as SPIGM(1,N), where N represents the number of variables and M is the training data length.

With the progressive incorporation of newly acquired data, the updating of the comprehensive system description facilitates more accurate characterization and instantaneous condition assessment. This iterative refinement process leverages accumulated measurement information to enhance the precision of prediction models, thereby enabling more reliable monitoring of the system and more informed decision-making regarding management strategies.

After using the difference operator, the equation

$$y^{(0)}(k) + a(k-M)z_1(k) = b_1(k-M)x_1^{(1)}(k) + b_2(k-M)x_2^{(1)}(k) + \dots + b_N(k-M)x_N^{(1)}(k) \quad (4)$$

is discrete form of SPIGM(1,N), where Z_1 be the adjacent mean generation sequence, and $z_1(k) = \frac{1}{2}(y^{(1)}(k) + y^{(1)}(k-1))$.

Similar to the traditional model, parameters are estimated in the next step. To control the computational burden, parameter estimation is carried out iteratively. Assume that $Y^{(0)}, X_i^{(0)}, Z_1$ are defined as above, the parameters of the training sample $\hat{\theta}$ are estimated to satisfy

$$\hat{\theta}_0 = (B_0^T B_0)^{-1} B_0^T Y_0 \quad (5)$$

where

$$B_0 = \begin{bmatrix} -z_1(2) & x_1^{(1)}(2) & x_2^{(1)}(2) & \dots & x_N^{(1)}(2) \\ -z_1(3) & x_1^{(1)}(3) & x_2^{(1)}(3) & \dots & x_N^{(1)}(3) \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ -z_1(M) & x_1^{(1)}(M) & x_2^{(1)}(M) & \dots & x_N^{(1)}(M) \end{bmatrix}, Y_0 = \begin{bmatrix} y^{(0)}(2) \\ y^{(0)}(3) \\ \vdots \\ y^{(0)}(M) \end{bmatrix}, \theta_0 = \begin{bmatrix} a(0) \\ b_1(0) \\ b_2(0) \\ \vdots \\ b_N(0) \end{bmatrix} \quad (6)$$

The parameters of the prediction stage $\hat{\theta}_k, k = M+1, M+2, \dots, M+m$ satisfies

$$\hat{\theta}_{k+1} = \begin{bmatrix} \hat{\theta}_k - \frac{(B_k^T B_k)^{-1} B_k^T h(k+1)h(k+1)^T P_k^\perp Y_k}{\frac{h(k+1)^T P_k^\perp h(k+1)}{h(k+1)^T P_k^\perp Y_k}} \\ \frac{h(k+1)^T P_k^\perp Y_k}{h(k+1)^T P_k^\perp h(k+1)} \end{bmatrix}, \quad (7)$$

where

$$B_k = \begin{bmatrix} -z_1(2) & x_1^{(1)}(2) & x_2^{(1)}(2) & \dots & x_N^{(1)}(2) \\ -z_1(3) & x_1^{(1)}(3) & x_2^{(1)}(3) & \dots & x_N^{(1)}(3) \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ -z_1(k) & x_1^{(1)}(k) & x_2^{(1)}(k) & \dots & x_N^{(1)}(k) \end{bmatrix}, Y_k = \begin{bmatrix} x_1^{(0)}(2) \\ x_1^{(0)}(3) \\ \vdots \\ x_1^{(0)}(k) \end{bmatrix}, \theta_k = \begin{bmatrix} a(k-M) \\ b_1(k-M) \\ b_2(k-M) \\ \vdots \\ b_N(k-M) \end{bmatrix} \quad (8)$$

and

$$h(k+1) = \begin{bmatrix} -z_1(k+1) & x_1^{(1)}(k+1) & x_2^{(1)}(k+1) & \dots & x_N^{(1)}(k+1) \end{bmatrix} \quad (9)$$

$$P_k^\perp = I - B_k (B_k^T B_k)^{-1} B_k^T \quad (10)$$

Further, to avoid computing the inverse matrix of $B_k^T B_k$, let

$$D_k = (B_k^T B_k)^{-1} \quad (11)$$

and using the recursion

$$D_{k+1} = \begin{bmatrix} D_k + \frac{D_k B_k^T h(k+1)h(k+1)^T B_k D_k}{h(k+1)^T P_k^- h(k+1)} & -\frac{D_k B_k^T h(k+1)}{h(k+1)^T P_k^- h(k+1)} \\ -\frac{h(k+1)^T B_k D_k}{h(k+1)^T P_k^- h(k+1)} & \frac{1}{h(k+1)^T P_k^- h(k+1)} \end{bmatrix} \quad (12)$$

where $P_k^- = I - B_k D_k B_k^T$. The minimum least square error is updated by using

$$J_{\min_{k+1}} = J_{\min_k} - \frac{(h(k+1)^T P_k^- Y_k)^2}{h(k+1)^T P_k^- h(k+1)}. \quad (13)$$

Then, the time recursion of SPIGM(1,N) is

$$\hat{y}^{(1)}(k) = \left[y^{(0)}(1) - \frac{1}{a(i)} \sum_{n=1}^N \hat{b}_n(i) \hat{x}_n^{(1)}(k) \right] e^{-\hat{a}(i)(k-1)} + \frac{1}{a(i)} \sum_{n=1}^N \hat{b}_n(i) \hat{x}_n^{(1)}(k) \quad (14)$$

and the prediction of $Y^{(0)}$ is

$$\hat{y}^{(0)}(k) = \hat{y}^{(1)}(k) - \hat{y}^{(1)}(k-1) \quad (15)$$

The modeling process and the parameter iteration are shown in Fig. 2 and its pseudocode is in Algorithm 1.

Algorithm 1: Iterative SPIGM prediction and update

Input: Training sequence $Y_{init} = \{y^{(0)}(1), \dots, y^{(0)}(k)\}$;

Output: Predictions $\hat{Y}_{pred}$, Estimated parameters Θ ;

Initialization: $Y_{current} \leftarrow Y_{init}$, $\hat{Y}_{pred} \leftarrow []$, $\Theta \leftarrow []$;

for $j = k$ **to** $T - 1$ **do**

1-order accumulated generation: $Y^{(1)} \leftarrow \text{AGO}(Y_{current})$

Estimate parameters: $\hat{\theta} \leftarrow (B^T B)^{-1} B^T Y_n$

Store current parameters: $\Theta.\text{add}(\hat{\theta})$

$\hat{y}^{(1)}(j+1) \leftarrow \left[y^{(0)}(1) - \frac{1}{a(i)} \sum_{n=1}^N \hat{b}_n(i) \hat{x}_n^{(1)}(k) \right] e^{-\hat{a}(i)(k-1)} + \frac{1}{a(i)} \sum_{n=1}^N \hat{b}_n(i) \hat{x}_n^{(1)}(k)$;

Prediction $\hat{y}^{(0)}(j+1) \leftarrow \hat{y}^{(1)}(j+1) - y^{(1)}(j)$

$\hat{Y}_{pred}.\text{add}(\hat{y}^{(0)}(j+1))$

update $y_{actual_next} \leftarrow Y_{new}[j-k+1]$, $Y_{current}.\text{add}(y_{actual_next})$;

return $\hat{Y}_{pred}$, Θ ;

2.3. Grey particle filtering

The idea of PF is based on Monte Carlo methods, which use particle sets to represent probability and can be used in any form of state space model. The state equation of grey particle filtering is derived from modeling process of SPIGM(1,N). Specifically, state equation

$$y(k) = SPIGM(y(k-1)) + w(k-1) \quad (16)$$

observation equation

$$y'(k) = y(k-1) + v(k-1) \quad (17)$$

where $w(k)$, $v(k)$ are Gaussian noise.

Generally, the process of grey particle filtering mainly includes the initialization, prediction, correction, resampling and filtering process. Grey particle filtering aims to find a set of random samples propagating under grey modeling process approximately present the probability density function. The minimum variance estimate of the system state is obtained by obtaining the sample mean. The pseudocode is in Algorithm 2 and specific steps are as follows:

Step 1: Initialization. A set of particles is generated according to prior knowledge or random sampling methods, each of which represents the possible state of the system and is given the same weight.

Step 2: Prediction. According to the system model Eq. (16), the state of each particle is predicted to obtain a particle sample at the next moment. When a priori distribution density is $p(\hat{y}_k^j | \hat{y}_{k-1}^j)$,

$$\hat{y}_k^j \sim p(\hat{y}_k^j | \hat{y}_{k-1}^j) \quad (18)$$

Step 3: Correction. According to the observation model Eq. (17) and the actual observation data, the weight of each particle w_k^j is calculated to evaluate the consistency of each particle with the current observation data. And normalization is carried out

$$\tilde{w}_k^j = \frac{w_k^j}{\sum_{i=1}^{N_p} w_k^i} \quad (19)$$

Step 4: Resampling. According to the weight of particles, retain high-weight particles and delete low-weight particles.

Step 5: Filtering. Repeat the above steps until the termination condition is reached, such as the set number of iterations or the estimated error reaches the predetermined threshold.

2.4. Model evaluation

Choose three error test standards to evaluate the accuracy of models. The mean absolute percentage error (MAPE) is the most important criterion for evaluating error. Moreover, standard deviation (STD) and root mean square percentage error (RMSPE) are also given. The detailed definitions are as follows:

(1) MAPE

$$MAPE = \frac{1}{n} \sum_{k=1}^n \left| \frac{\hat{y}^{(0)}(k) - y^{(0)}(k)}{y^{(0)}(k)} \right| \times 100\% \quad (20)$$

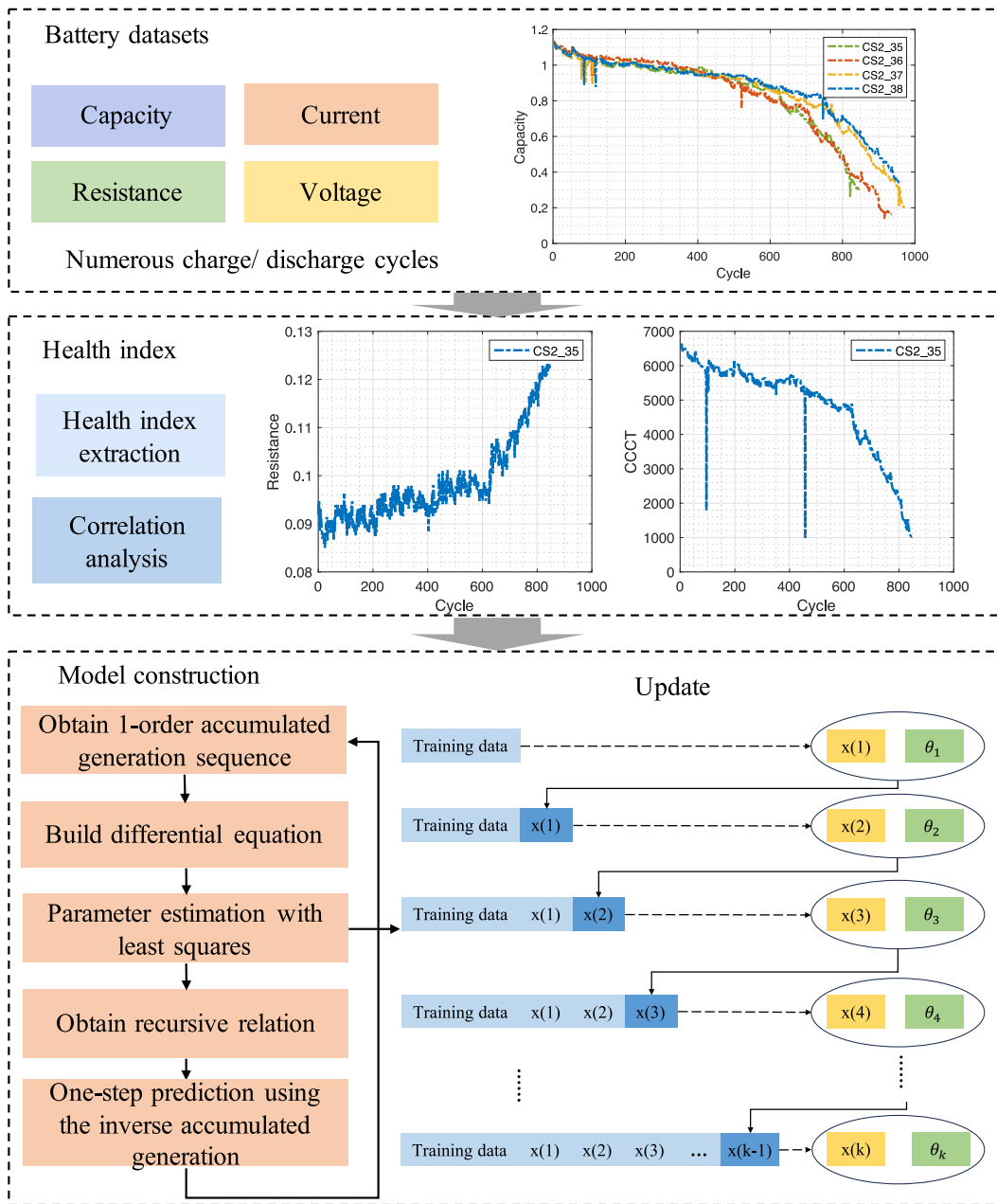


Fig. 2. Modeling process for SPIGM(1,N).

(2) STD

$$STD = \sqrt{\frac{1}{n} \sum_{k=1}^n \left(\left| \frac{\hat{y}^{(0)}(k) - y^{(0)}(k)}{y^{(0)}(k)} \right| - MAPE \right)^2} \quad (21)$$

(3) RMSPE

$$RMSPE = \sqrt{\frac{1}{n} \sum_{k=1}^n \left(\frac{\hat{y}^{(0)}(k) - y^{(0)}(k)}{y^{(0)}(k)} \right)^2} \times 100\% \quad (22)$$

Algorithm 2: Grey particle filtering for battery capacity prediction

Input: Number of particles N_p , capacity measurements $\{y'_{n+1:n+T}\}$, process noise covariance Q , measurement noise covariance R , initial capacity $\{y_1, \dots, y_n\}$

Output: Prediction $\{\hat{y}_k\}_{k=n+1}^{n+T}$

```

for  $i = 1$  to  $N_p$  do
  | Sample initial capacity:  $y_n^i \sim \mathcal{N}(y_n, \sigma_0^2)$ ; Initialize history buffer:  $H_n^i \leftarrow (y_1, y_2, \dots, y_{n-1}, y_n^i)$ ; Set initial weight:  $w_n^i \leftarrow 1/N_p$ ;
end

for  $k = n + 1$  to  $n + T$  do
  for  $i = 1$  to  $N_p$  do
    | Extract history:  $Y^{(0)} \leftarrow H_{k-1}^i = (y_{k-n}^i, \dots, y_{k-1}^i)$ ; Predict mean capacity:  $\mu_k^i \leftarrow \text{SPIGM}(Y^{(0)})$ ; Sample from process model:
    |  $y_k^i \sim \mathcal{N}(\mu_k^i, Q)$ ; Update history buffer:  $H_k^i \leftarrow (y_{k-n+1}^i, \dots, y_{k-1}^i, y_k^i)$ ;
  end

  Obtain measurement:  $y'_k$ ;
  for  $i = 1$  to  $N_p$  do
    | Compute likelihood weight:  $w_k^i \leftarrow \exp\left(-\frac{(y'_k - y_k^i)^2}{2R}\right)$ ;
  end

   $W_{total} \leftarrow \sum_{j=1}^{N_p} w_k^j$ ;
  for  $i = 1$  to  $N_p$  do
    |  $w_k^i \leftarrow w_k^i / W_{total}$ ;
  end

   $\hat{y}_k \leftarrow \sum_{i=1}^{N_p} w_k^i \cdot y_k^i$ ;

  Compute effective sample size:  $N_{eff} \leftarrow 1 / \sum_{i=1}^{N_p} (w_k^i)^2$ ;
  if  $N_{eff} < N_p / 2$  then
    | Resample particles:  $\{y_k^i, H_k^i\}_{i=1}^{N_p} \leftarrow \text{Resample}(\{y_k^i, H_k^i, w_k^i\}_{i=1}^{N_p})$ ; Reset weights:  $w_k^i \leftarrow 1/N_p$  for all  $i$ ;
  end
end

return  $\{\hat{y}_k\}_{k=n+1}^{n+T}$ ;

```

3. Predictive maintenance strategy

Only through reasonable and efficient maintenance strategies can normal and safe equipment operation be ensured. In PHM, specific failure thresholds are typically established to evaluate equipment health status. For instance, in BMS 70% and 80% SOH are commonly employed as failure thresholds. Since equipment failure results in significant downtime losses, predictive maintenance during operational periods is of paramount importance.

Under the framework of the classic maintenance strategy, maintenance is often conducted in equal time intervals because its execution time is clear. Assuming that the current cycle is h and maintenance are conducted after the interval cycles T , the maintenance is performed in

$$TBM = \begin{cases} 1, h = NT \\ 0, else, \end{cases} \quad (23)$$

this maintenance strategy is called time-based maintenance (TBM).

Then, when the state of a system can be obtained, maintenance is implemented that depends on the state change. Assuming maintenance is not conducted at the current use cycle h , the possibility of failure is

$$P(State \leq \eta) \quad (24)$$

where η is a failure threshold. The maintenance is executed in

$$CBM = \begin{cases} 1, P(State \leq \eta) \geq \Phi \\ 0, P(State \leq \eta) \leq \Phi \end{cases} \quad (25)$$

where Φ represents a predefined threshold value used to determine whether maintenance operations should be executed. This kind of maintenance strategy is called condition-based maintenance (CBM).

However, there are key parameters that need to be set in TBM and CBM. In TBM, setting different T has a great impact on the maintenance cost rate. A smaller T will cause excessive maintenance, and a larger T will cause a possibility of failure. In CBM,

the value of Φ brings different optimization decision-making results. In such a situation, it is important to design a predictive maintenance strategy without parameter setting.

Due to the special nature of lithium batteries, they will not fail suddenly like traditional mechanical equipment. Battery capacity is a process of gradual decline. The execution of maintenance behavior will not return the battery capacity to the rated state. Therefore, only one maintenance is considered here. Next, the proposed maintenance is designed combined TBM with CBM.

Assuming C_T is the cost of maintenance in TBM, the cost rate is defined as

$$CR_{TBM} = \frac{C_T}{h} \quad (26)$$

where h represents the current cycle when maintenance is implemented. Obviously, CR_{TBM} is a function that decreases with the increase of h .

Assuming C_C is the cost that includes downtime cost in CBM. According to the method proposed in Section 2, the specific failure probability value at each cycle can be calculated. The threshold of failure probability can be ignored. Once the maintenance is executed, it means that there is a risk of failure in the next cycle $h + 1$. $State_h$ is the health state in the cycle h . Thus, the cost rate for the decision is

$$CR_{CBM} = \frac{C_C \times P(State_h \leq \eta)}{h + 1} = \frac{C_C}{h + 1} \int_{-\infty}^{\eta} f(\tau) d\tau \quad (27)$$

Obviously, CR_{CBM} is a function that increases with the increase of $P(State_h \leq \eta)$.

Then, the best cycle will be found according to CR_{TBM} and CR_{CBM} , i.e.

$$h^* = \left\{ h \mid \frac{C_T}{h} = \frac{C_C}{h + 1} \int_{-\infty}^{\eta} f(\tau) d\tau \right\} \quad (28)$$

Some underlying assumptions are made in the maintenance strategy:

- (1) If the failure occurs, carry out the maintenance work immediately.
- (2) If $CR_{TBM} < CR_{CBM}$, carry out the time-based maintenance.
- (3) Considering the downtime loss, assume that $C_T < C_C$.

For maintenance decisions, the maintenance cost per unit cycle is used as an evaluation indicator. It is defined as

$$MCPU = \frac{C_T}{h^*} \times I(h^* \leq t_{EOL}) + \frac{C_C}{t_{EOL}} \times I(h^* > t_{EOL}) \quad (29)$$

where h^* is the planned maintenance cycle and t_{EOL} is the time when the device reaches the end of life (EOL). MCPU serves as an indicator of the economic performance of various maintenance strategies. Lower MCPU values indicate more cost-effective approaches, making such strategies more economically favorable.

4. Case study

4.1. Dataset description

The dataset was obtained from the University of Maryland Advanced Life-Cycle Engineering Center [48]. All CS2 cells were subjected to identical charging protocols following the standard constant current/constant voltage (CC-CV) mode. The cells were charged at a constant current rate of 0.5C until the terminal voltage reached 4.2 V, then maintained at 4.2 V until the charging current decreased below 0.05 A. The rated capacity of these cells is 1.1 Ah. For this study, the four battery cells (CS2_35, CS2_36, CS2_37 and CS2_38) in this dataset were selected because they represent typical battery degradation behavior under common operating conditions. These batteries are cycled using CC-CV charging protocol, the most widely adopted standard in lithium-ion battery applications, resulting in representative degradation trajectories. Other datasets present specific degradation characteristics, such as NASA (regeneration phenomena) and MIT (capacity knee points). To validate the applicability to common battery degradation prediction scenarios, the CS2 dataset with its conventional degradation behavior was chosen. It demonstrates the effectiveness of the proposed method across varying degradation stages and different prediction starting points.

4.2. Correlation analysis

Numerous factors, including electrochemical aging, thermal stress, mechanical degradation, and operational conditions influence battery capacity degradation. Most health indicators are extracted from current and voltage profiles during charging and discharging cycles. These profiles contain rich information about the battery internal state and provide valuable insights into degradation mechanisms. In this study, constant current charge time (CCCT) and internal resistance are selected as health factors for battery capacity prediction. CCCT represents the duration required to charge the battery at a constant current rate until the voltage reaches the upper cutoff limit, serving as an indicator of the ability to accept charge. Internal resistance reflects the impedance characteristics of the battery and typically increases as the battery ages due to active material loss and electrolyte decomposition.

Prior to prediction model development, correlation analysis is conducted on the selected health factors to validate their relationship with battery capacity. Fig. 3 illustrates the relationship between capacity and the chosen health factors, demonstrating

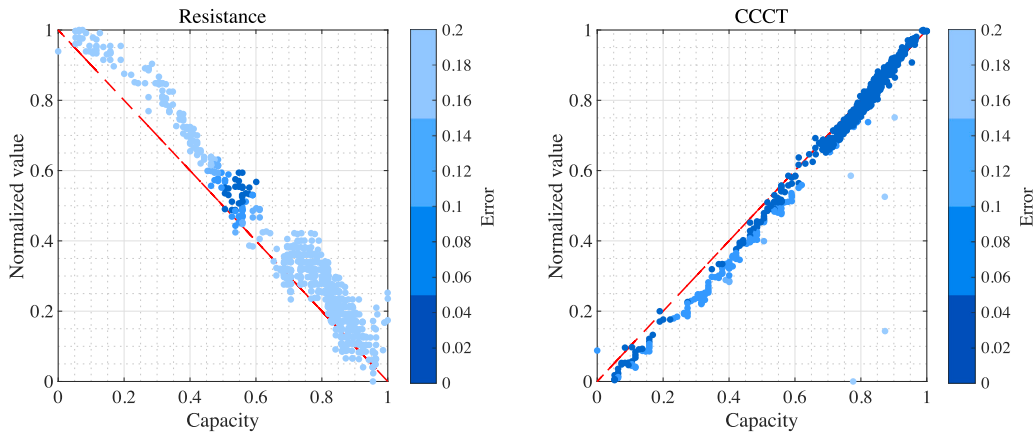


Fig. 3. Relation between capacity and health factors.

Table 1
Correlation analysis between capacity and health factors.

	CS2_35	CS2_36	CS2_37	CS2_38
Pearson correlation coefficient				
Resistance	−0.9780	−0.9771	−0.9792	−0.9794
(<i>p</i> -value)	< 0.001	< 0.001	< 0.001	< 0.001
CCCT	0.9831	0.9755	0.9792	0.9797
(<i>p</i> -value)	< 0.001	< 0.001	< 0.001	< 0.001
Grey correlation degree				
Resistance	0.7132	0.7198	0.7235	0.7043
CCCT	0.9298	0.9326	0.9344	0.9201

clear trends that support their use as predictive variables. Table 1 presents the detailed correlation analysis results. Generally, Pearson correlation coefficient greater than 0.8 and a grey correlation coefficient greater than 0.7 indicate strong correlation between variables. As shown in Table 1, the selected health factors satisfy these criteria and are therefore suitable for multivariable prediction modeling. The analysis reveals that internal resistance exhibits a negative correlation with capacity. Conversely, CCCT demonstrates a positive correlation with capacity, suggesting that longer charging times at constant current are associated with higher remaining capacity.

4.3. Capacity prediction

Fig. 4 presents the capacity prediction results for the four batteries using the proposed model along with their corresponding absolute percentage errors (APE). Specifically, the prediction starting point is set at 70% of the total cycle life. The initial model is constructed using the first 70% of the actual capacity data. This model is then used to forecast the capacity at the subsequent cycle $k + 1$. When the actual capacity measurement for cycle $k + 1$ becomes available, it first serves as the ground truth to evaluate the accuracy of the forecast made at cycle k . Subsequently, this observed data point is assimilated into the historical dataset to update the model parameters for forecasting the next cycle $k + 2$. This iterative process continues until the EOL. Table 2 provides a comprehensive accuracy evaluation of the SPIGM(1,N) performance. As demonstrated in Fig. 4, SPIGM exhibits highly satisfactory predictive performance for battery capacity forecasting across all four test cases. The model successfully captures the overall degradation trends with remarkable precision. Excluding occasional noise-induced deviations, the prediction accuracy consistently remains below 3%. It demonstrates robust performance under varying operational conditions. Despite the different cycle lengths among the four batteries, the prediction results maintain consistent quality across various starting points. The accuracy evaluation presented in Table 2 reveals that during the model training phase, the MAPE of SPIGM is less than 1.5%, which demonstrates the rationality and effectiveness of SPIGM for the application in capacity degradation prediction. During the prediction phase, although MAPE values increase as expected due to the inherent uncertainty in long-term forecasting, the model continues to deliver relatively excellent predictive performance. The consistent performance across batteries with varying degradation characteristics and cycle histories further confirms the robustness and generalizability of the SPIGM(1,N) model for real-world battery capacity prediction scenarios.

Further, to show the superiority of SPIGM(1,N), some models have been selected as comparison models. Grey prediction models include GM(1,N), FGM(1,N), MGM(1,N), CFGM(1,1) [49], ADGM(1,1) [50] and AVFGM(1,N) [51]. Traditional machine learning methods include SVR, RF and XGBoost. Deep learning methods include CNN, MLP and Transformer. Hybrid methods include GPR-LSTM and CNN-LSTM [52]. Table 3 lists the accuracy evaluation comparison of different models.

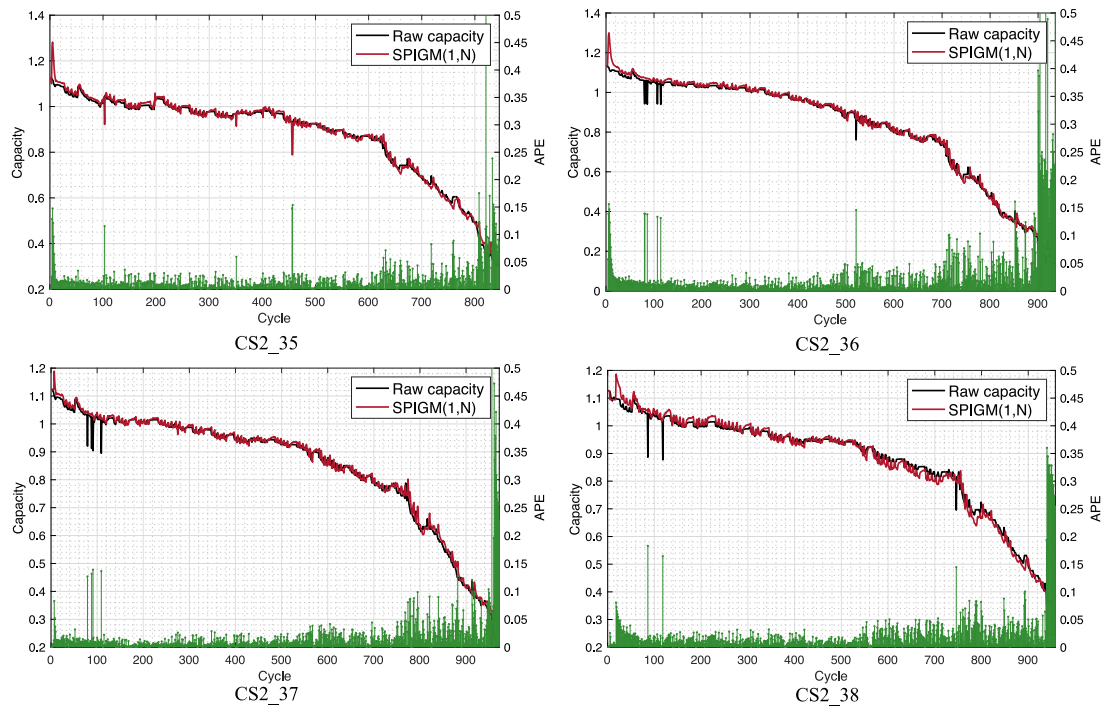


Fig. 4. Capacity prediction results.

Table 2
Accuracy evaluation of proposed SPIGM(1,N).

		MAPE (%)	RMSPE (%)	STD (%)
CS2_35	training	1.1966	1.9469	1.1847
	prediction	1.2551	3.8196	1.2229
CS2_36	training	1.4454	2.4212	1.4311
	prediction	2.6302	4.2247	2.5745
CS2_37	training	1.0070	1.6141	0.9970
	prediction	2.2704	2.3613	2.5745
CS2_38	training	1.3726	1.9714	1.3589
	prediction	2.7120	4.7642	2.6655

Table 3
Accuracy evaluation (MAPE) comparison.

	CS2_35	CS2_36	CS2_37	CS2_38
GM(1,N)	5.26	7.40	6.71	7.63
FGM(1,N)	9.91	6.69	7.03	3.83
MGM(1,N)	17.61	4.07	13.48	10.57
CFG(1,1)	21.05	11.26	18.40	11.94
ADGM(1,1)	25.54	10.39	17.45	20.61
AVFGM(1,N)	1.70	2.53	1.98	2.03
SVR	9.71	11.39	11.06	7.59
RF	9.93	15.91	10.24	7.67
XGBoost	3.33	6.20	5.03	5.63
CNN	8.51	16.97	17.01	13.44
MLP	11.30	6.81	3.59	2.71
Transformer	1.65	4.61	3.25	2.96
GPR-LSTM	5.92	4.19	2.47	2.32
CNN-LSTM	2.69	6.70	5.76	3.92
SPIGM	1.26	2.63	2.27	2.71

It is noted from Table 3 that while the proposed SPIGM model achieves the best average performance, other models, particularly AVFGM, occasionally yield higher accuracy on individual cells (e.g., CS2_38). We attribute this phenomenon to fundamental differences in the model structures. AVFGM is a multi-variable model that incorporates more health factor variables into its

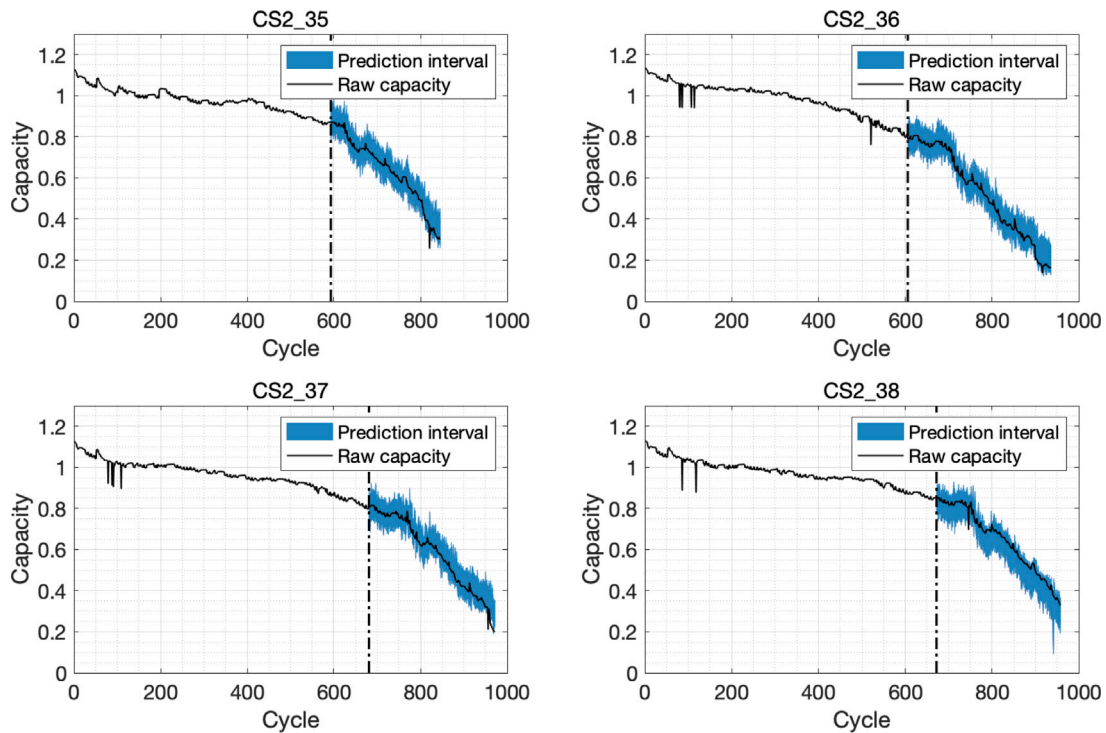


Fig. 5. Capacity probability prediction for batteries.

prediction. And AVFGM contains two nonlinear parameters. The joint action of these parameters provides the model with significantly greater flexibility, which allows it to fit the specific degradation trajectories of certain individual cells more closely. While this complexity benefits AVFGM in specific instances, our proposed SPIGM model demonstrates superior overall robustness and average accuracy across all test cases while requiring fewer inputs.

These superior performances can be attributed to several key points. First, SPIGM outperforms traditional grey prediction models due to its continuous iteration of both structure and parameters, enabling the model to better align with actual degradation characteristics through adaptive learning. Second, SPIGM demonstrates competitive performance compared to machine learning approaches. In the comparative analysis, machine learning models were implemented using default hyperparameter configurations, resulting in prediction accuracies lower than those achieved by SPIGM. While we acknowledge that systematic hyperparameter optimization could potentially improve machine learning model performance, this process typically requires extensive computational resources and domain expertise for parameter tuning. The proposed SPIGM model has the notable advantage that it eliminates the need for complex parameter optimization procedures. This characteristic significantly enhances the SPIGM practical utility which makes it particularly attractive for real-world battery management applications where simplicity, reliability, and ease of deployment are critical considerations.

4.4. Predictive maintenance evaluation

The effectiveness of the proposed method in battery degradation prediction has been verified in Section 4.3. In this section, the probability prediction will be carried out in combination with the grey filtering algorithm.

Considering the inherent error of battery capacity and resistance measurement, when using grey filtering to update and predict, Gaussian noise is given to measure the uncertainty of the measurement. For the generality of uncertainty, experiments are carried out under the condition of 50 random noise matrix values. The probability prediction results of the capacity are shown in Fig. 5. It can be seen from Fig. 5 that due to the influence of noise, there is a large distance between the upper and lower bounds of the prediction interval. Except for some anomalies, the prediction results as a whole revolve around a single SPIGM prediction result.

Fig. 6 shows the probability distribution of results at prediction steps and Fig. 7 shows the prediction uncertainty. The peak of probability density represents the most likely situation of the prediction results obtained after taking into account the uncertainty of the measurement by the proposed method. On the whole, the distribution of the predicted results is consistent with the actual real value. Notably, for the CS2_35 battery, the peak of its predicted probability density fell below the failure threshold at an early prediction step in Fig. 7. This is attributable to a significant and rapid degradation phase in its original data. Consequently, the model learned and extrapolated this higher degradation rate, resulting in a higher predicted probability of failure early on. For instance, at the 60th prediction step, the predicted probability of the capacity falling below the failure threshold had reached 90%.

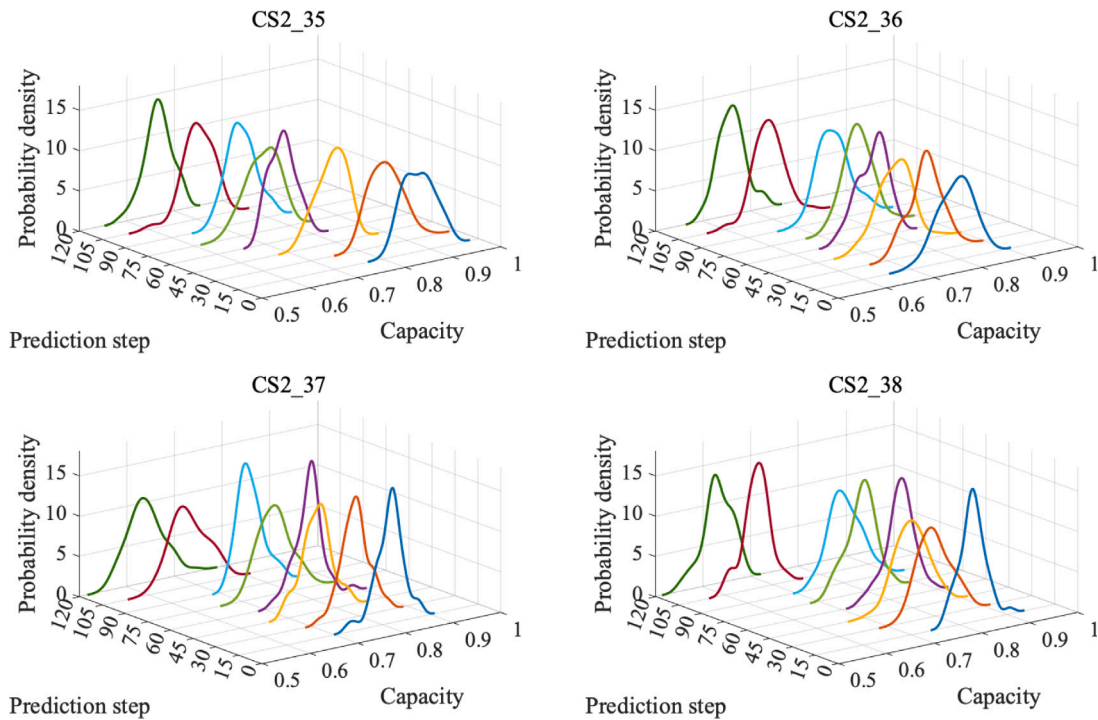


Fig. 6. The probability of results at different prediction steps.

Although the result of CS2_35 is relatively large, the peak probability density of CS2_36, CS2_37 and CS2_38 is close to the real value. This fact also shows the effectiveness of the probabilistic prediction method proposed in this paper.

Next, we will design the maintenance strategy according to the probability prediction results. According to the basic assumptions, the economic level of maintenance strategy is evaluated by the cost rate. The battery failure threshold η is set to 0.77Ah (70% of the rated capacity).

Fig. 8 shows the cost rate changes for different maintenance strategies. It can be seen that the cost rate of TBM shows a decreasing trend. As a basic common sense, the shorter the use time is, the higher the cost rate is. With the increase in use time, as long as the failure does not occur, the cost rate will gradually decrease and its economic level will become higher. For CBM, perform maintenance according to the health status of the battery. In the early stage of the prediction, there is no possibility of reaching the failure threshold in the prediction results. Therefore, due to the absence of maintenance actions, the cost rate is 0. As the possibility of prediction results that reach the failure threshold increases, the cost rate gradually increases.

Under the trend change of the cost rate curve of TBM and CBM, an interesting point occurs, that is, the intersection of the two curves. More interestingly, this point occurs before the failure threshold. Hence, this point where this intersection is located will be defined as the planned maintenance cycle. It can be seen from Fig. 8 that there are many intersections. Here, we take the time of the first occurrence as the planned maintenance cycle. Then, the proposed strategy is that adopt the CBM maintenance strategy before the planned cycle, and the TBM maintenance strategy after the planned cycle. Such a hybrid maintenance strategy greatly reduces the maintenance cost rate and improves the economic level of the battery.

4.5. Sensitivity analysis

The maintenance strategy designed according to the capacity probability prediction takes into account the health state of different cycle stages. To further discuss the sensitivity analysis of the proposed maintenance strategy to different maintenance costs, set different parameters to show its economic level. Table 4 lists the change of the planned maintenance cycle with the change of the proportion of C_C/C_T . With 70% of the rated capacity as the failure threshold, the EOL of the four batteries is also listed in Table 4.

It can be seen from Table 4 that with the increase of C_C/C_T , the planned maintenance cycles are constantly ahead of schedule. Compared with EOL, when C_C/C_T is larger, the planned maintenance cycle is farther away from EOL. This is precisely because the downtime loss after the failure is considered in C_C . When the downtime loss is large, it is safer to take maintenance action as soon as possible. On the other hand, when C_C and C_T are closer, the maintenance operation occurs later. Because when the cost is the same, there is not much need for early maintenance. This conclusion is completely consistent with the scenarios in practical engineering applications. Moreover, when C_C/C_T ratio increases beyond a certain threshold, the optimal maintenance cycle stabilizes and no longer changes significantly. This finding suggests that there exists a saturation point in the cost-benefit relationship. The further increases in the C_C relative to C_T have diminishing impact on maintenance strategy optimization.

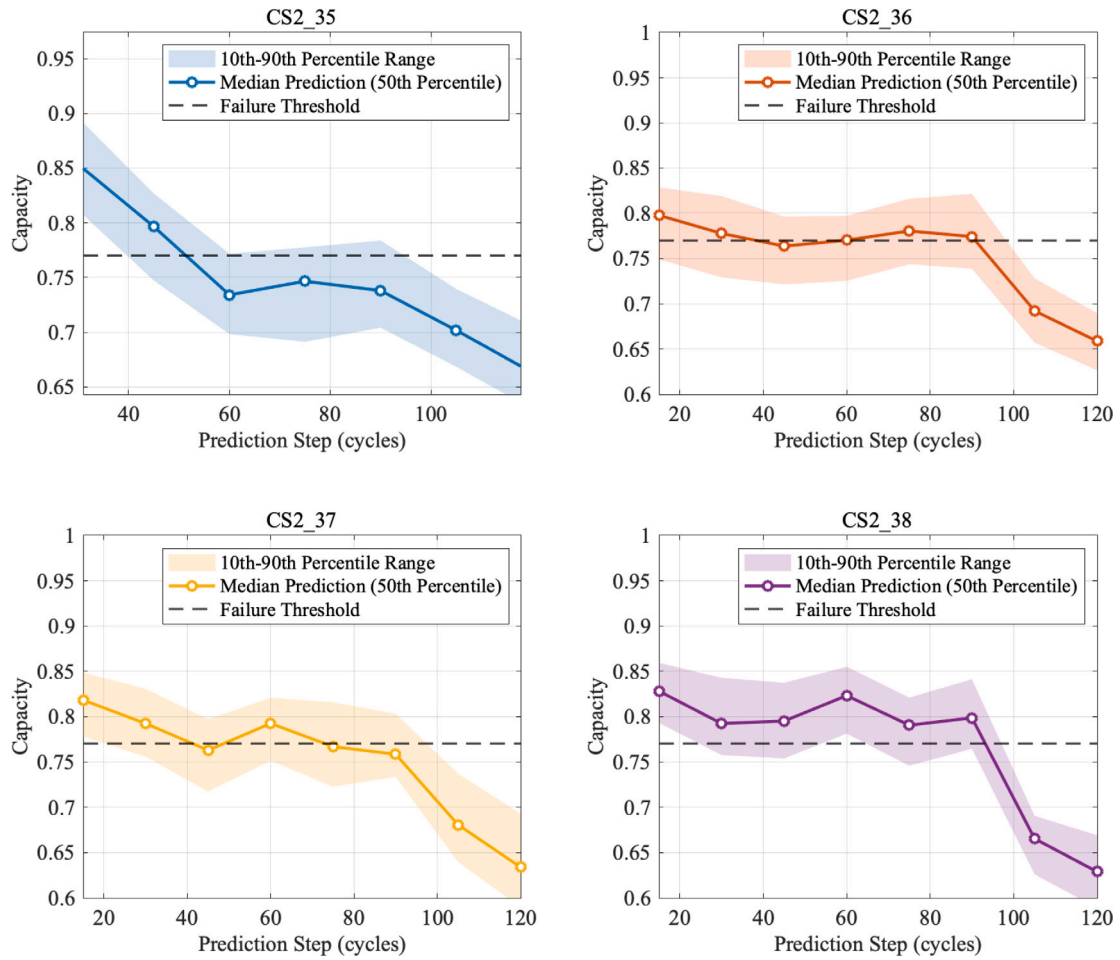


Fig. 7. Prediction with 80% prediction interval.

Table 4

Change of planned maintenance cycle.

C_C/C_T	1	2	3	4	5	6	EOL
CS2_35	493	471	470	467	467	467	639
CS2_36	523	427	424	420	420	420	637
CS2_37	580	513	507	491	490	490	712
CS2_38	564	544	516	499	499	499	754

Fig. 9 shows the comparison of MCPU for maintenance strategies. It can be seen from Fig. 9 that when C_C and C_T are the same value, the MCPU of the TBM maintenance strategy is lower than that of the proposed maintenance strategy. This shows that there is no need to worry about downtime loss after the failure at this time. It is the best decision to perform maintenance according to the use time of the battery. When C_C/C_T increases, the MCPU of the proposed maintenance strategy is much lower than that of TBM. However, when C_C/C_T increases, the MCPU of the proposed maintenance strategy does not increase significantly. This is because the changes in the planned maintenance cycles in Table 4 are also gradually decreasing. To a certain extent, it also shows the controllability and stability of the proposed method in terms of cost-effectiveness.

5. Conclusion

To predict battery degradation accurately and enhance maintenance timeliness, this paper proposes a battery management framework. The main contributions are as follows:

- (1) This study addresses battery degradation prediction and formulates an optimal predictive maintenance strategy based on prediction model results. It provides practical guidance for battery management systems.

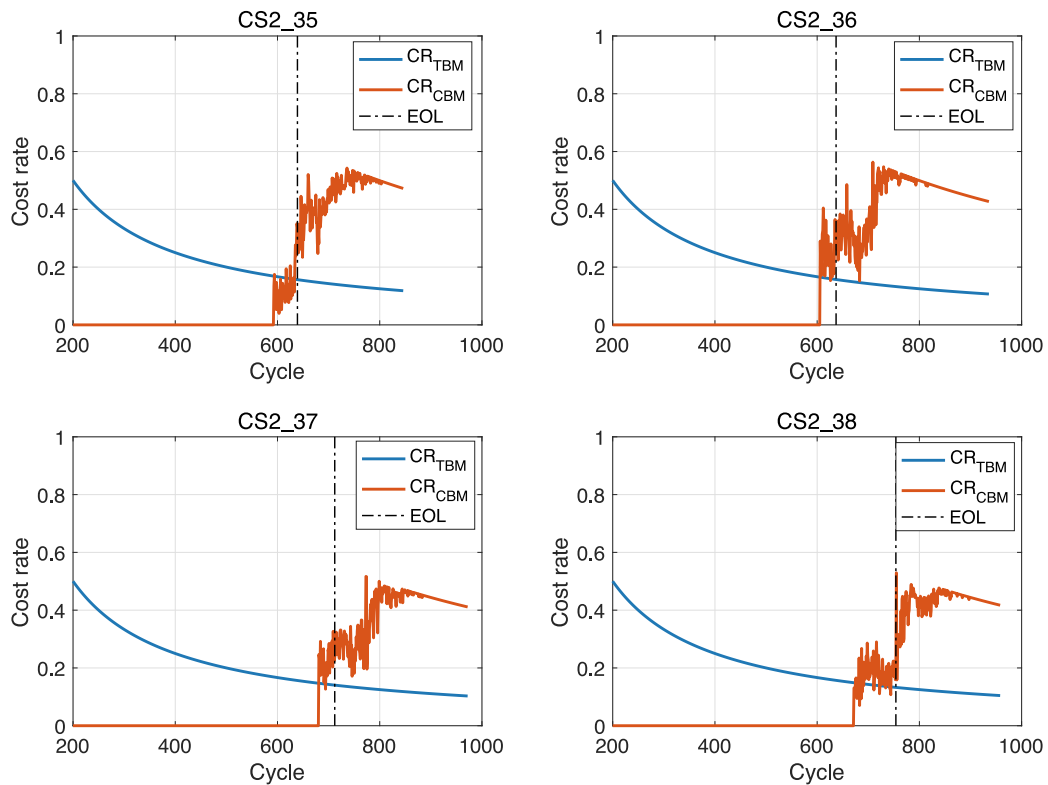


Fig. 8. Cost rate for maintenance strategy.

- (2) The prediction method accounts for the cumulative effects of degradation and measurement-induced errors. It integrates a deterministic prediction model with an iterative structure filtering algorithm. The designed predictive maintenance strategy incorporates maintenance cost rates in different health states based on probabilistic prediction.
- (3) Case study results demonstrate the superiority of the proposed approach. The prediction method achieves higher accuracy than popular existing models across different prediction starting points. Sensitivity analysis of the maintenance strategy further validates the cost-effectiveness of the proposed hybrid maintenance framework.

Although this research yields innovative results, several areas warrant further investigation. Future research should explore extension to multi-battery management, where the framework could account for heterogeneous degradation patterns under diverse conditions. Analysis of battery degradation curves reveals that batteries exhibit varying degradation rates across different operational stages. Developing multi-stage prediction and maintenance strategies that align with stage-specific degradation characteristics would better reflect actual degradation behavior and improve practical applicability. Additionally, integrating the grey model with physics-based models presents a promising hybrid approach, which could enhance both prediction accuracy and model interpretability.

CRediT authorship contribution statement

Kailing Li: Writing – original draft, Software, Methodology, Conceptualization. **Naiming Xie:** Writing – review & editing, Supervision, Funding acquisition. **Nan Chen:** Writing – review & editing, Validation, Supervision. **Wenjie Dong:** Writing – review & editing, Visualization. **Liudong Gu:** Writing – review & editing, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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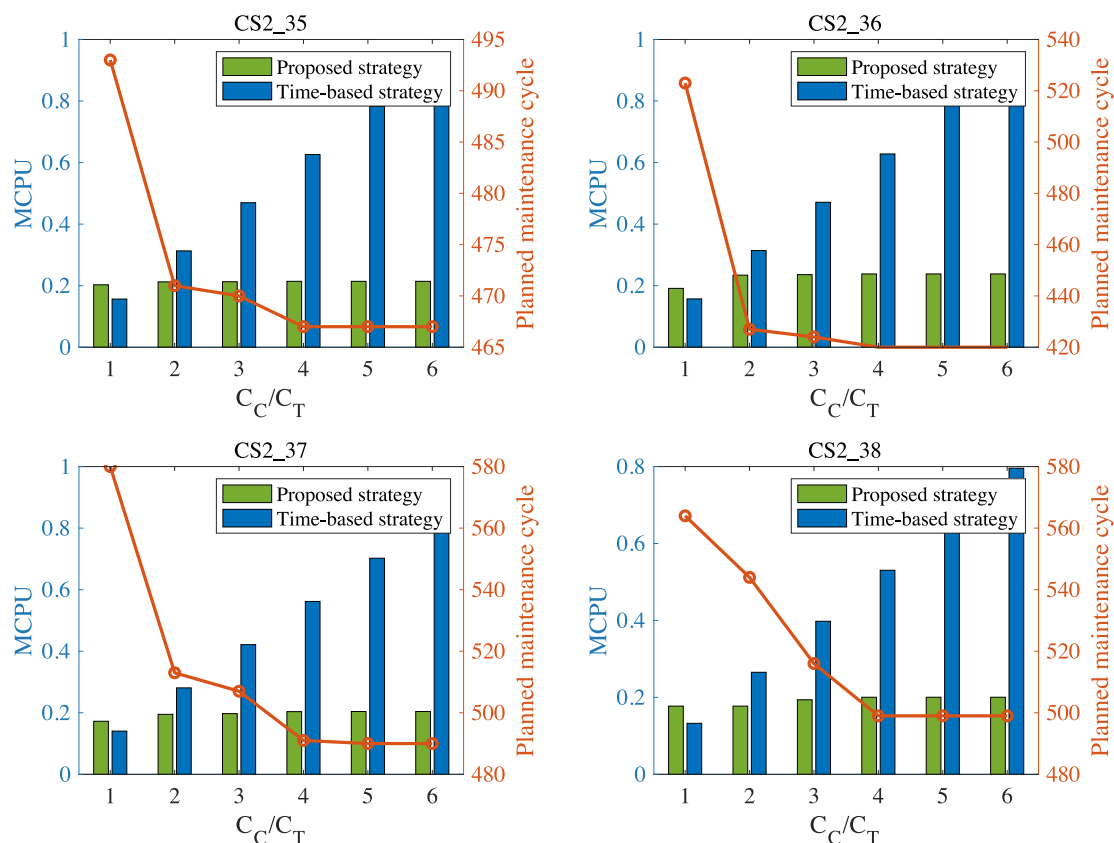


Fig. 9. MCPU for maintenance strategies.

Data availability

Data will be made available on request.

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